Da Xu
China Mobile Research Institute, Beijing, China
xudayj@chinamobile.com

30 October 2026

Editors
IEEE Transactions on Software Engineering

Dear Editors,

Please consider *Adaptive HACT: Information Limits and Safe Adaptation of Verification Evidence*, by Da Xu and Fei Wang, for review as a regular research article.

The paper studies how software-verification evidence can be reorganized without changing its acceptance contract. Its principal contribution is an explicit family of completion laws with identical exact singleton and pairwise summaries but disjoint globally optimal evidence trees, together with sharp minimax regret. A complete-epoch construction and fixed-eight candidate-class extension connect that information limit to batched check execution while keeping their distinct scopes explicit.

A complementary result gives an exact global baseline under exchangeable complete batches. We attribute the cardinality-state principle to established batch-update tree optimization and explain the model-specific specialization. The paper also gives a conditional local publication invariant and a paired-cost adoption rule that accounts for calibration, declared workload shift and installation.

The empirical contribution is a reproducible boundary study. Retrospective scoring preserves the symmetric baseline's item/file policy reversal and original-holdout losses, the maintained-counter ledger's service advantage, the original Packaging resource exclusion and six initial VOID runs. A separate sparse compact replay admits Packaging's complete 26,948-check registry within a 2 GiB budget; exclusive CPU profiles locate implementation work without replacing the unprofiled service comparison. The article makes no general service or agent-speedup claim. This combination of information limits, acceptance-preserving composition and independently checked workload boundaries is intended to inform the design of verification infrastructure for software agents.

The accompanying technical supplement provides full complete-epoch and padding proofs and an artifact guide. Code, exact references, independent audits and all retained observations are publicly available in the versioned artifact. AI assistance is disclosed in the manuscript and supplement. The submission metadata records the public repository versions.

Thank you for considering the work.

Sincerely,
Da Xu
Corresponding author for Da Xu and Fei Wang